Diamond Price Prediction

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Abstract

Diamond price prediction is treated as a supervised tabular regression problem in which the response is sale price in U.S. dollars and the predictors describe weight, physical dimensions, proportions, cut, color, and clarity. The study uses the standard diamonds data set, compares a mean baseline, log-target linear regression, Random Forest, XGBoost, and a multi-layer perceptron (MLP), and selects the final model by five-fold cross-validated RMSE on the training split.XGBoost was the best cross-validated model, with RMSE 524.50 ± 8.22 USD, MAE 261.78 ± 2.85 USD, and $R^2 = 0.9826 \pm 0.0004$. On the single held-out test evaluation, XGBoost achieved RMSE 534.80 USD, MAE 269.49 USD, $R^2 = 0.9825$, and MAPE 0.0628. The main finding is that nonlinear tree ensembles capture the dominant size-quality interactions substantially better than the linear baseline, while the completed MLP baseline is competitive but does not improve over XGBoost or Random Forest on this cleaned data set.

1 Introduction and Problem Description

Diamonds are priced from a combination of physical size, geometric proportions, and quality grades. The predictive task is a supervised regression problem: given a feature vector x containing carat, cut, color, clarity, depth, table, and dimensions x, y, z, estimate the continuous target p, the transaction price in U.S. dollars. The public diamonds data set is well suited for this task because it has a moderate tabular scale, a mixed numeric-categorical schema, and a strongly skewed monetary target [1].

The project objective is not only to obtain an accurate predictor, but also to compare model families with different inductive biases. A mean-price regressor defines the minimum reference. Linear regression tests whether log-price can be explained with mostly additive effects after scaling. Random Forest and XGBoost test tree ensembles that can model nonlinear thresholds and interactions. The final MLP extends the comparison to a neural-network baseline trained on scaled numeric features and one-hot categorical grades. This model set gives a pragmatic comparison between interpretable linear structure, high-performing tree ensembles, boosting, and neural regression.

2 Dataset Analysis

The source data set contains 53,940 rows and 10 variables: price, carat, cut, color, clarity, depth, table, and dimensions x, y, and z [1]. Price ranges from 326 to 18,823 USD, while carat ranges from 0.20 to 5.01. The categorical quality variables are ordered: cut ranges from Fair to Ideal, color from J to D, and clarity from I1 to IF. The project CSV includes the same schema after dropping any unnamed index column.

Data validation checks that all required columns are present, that there are no missing values, and that price is strictly positive. Cleaning removes physically invalid diamonds with non-positive dimensions, removes extreme dimensional outliers with $y \geq 30$ mm or $z \geq 30$ mm, and drops exact duplicates. This reduces the data from 53,940 to 53,772 observations. The cleaned data are split 80/20 using a fixed random seed and stratification on carat quintiles, producing 43,017 training rows and 10,755 held-out test rows.

Exploratory plots in Fig. 1 show two important properties. First, price is strongly right-skewed. Most diamonds are comparatively low priced, while a smaller number of high-value diamonds extend the upper tail. This supports the use of a log-transformed target during model fitting: it reduces the dominance of expensive diamonds during training while still allowing final errors to be reported in USD after inverse transformation. Second, carat and the physical dimensions form a size-related block that is strongly associated with price. The correlation heatmap shows that the size features are also collinear with one another. This collinearity is relevant for the linear model, where coefficients can become unstable, but it is less problematic for tree ensembles that can select among correlated split variables.

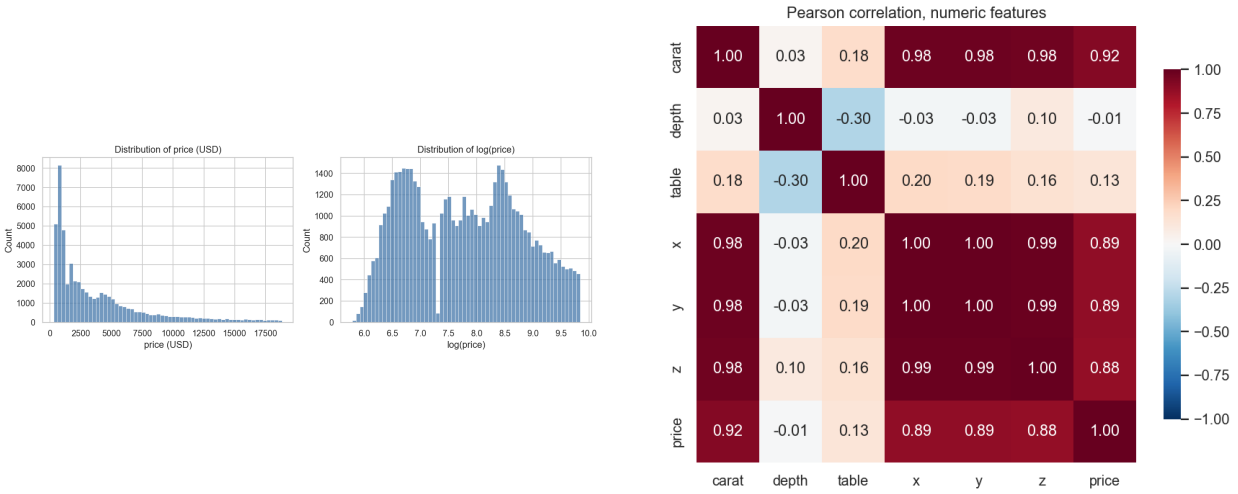


Figure 1: Exploratory data analysis. The target is right-skewed on the original dollar scale, while the log transform is more symmetric. The numeric correlation structure shows that carat and physical dimensions are a highly related size block.

Carat has the clearest visible relationship with price. The relationship is nonlinear: prices rise slowly for small stones, then increase more sharply for larger stones. The spread also increases as carat increases, indicating that quality grades and proportions matter more at higher price levels. This pattern is consistent with the later model results: a purely linear model captures the general trend, but tree ensembles capture nonlinear interactions more effectively.

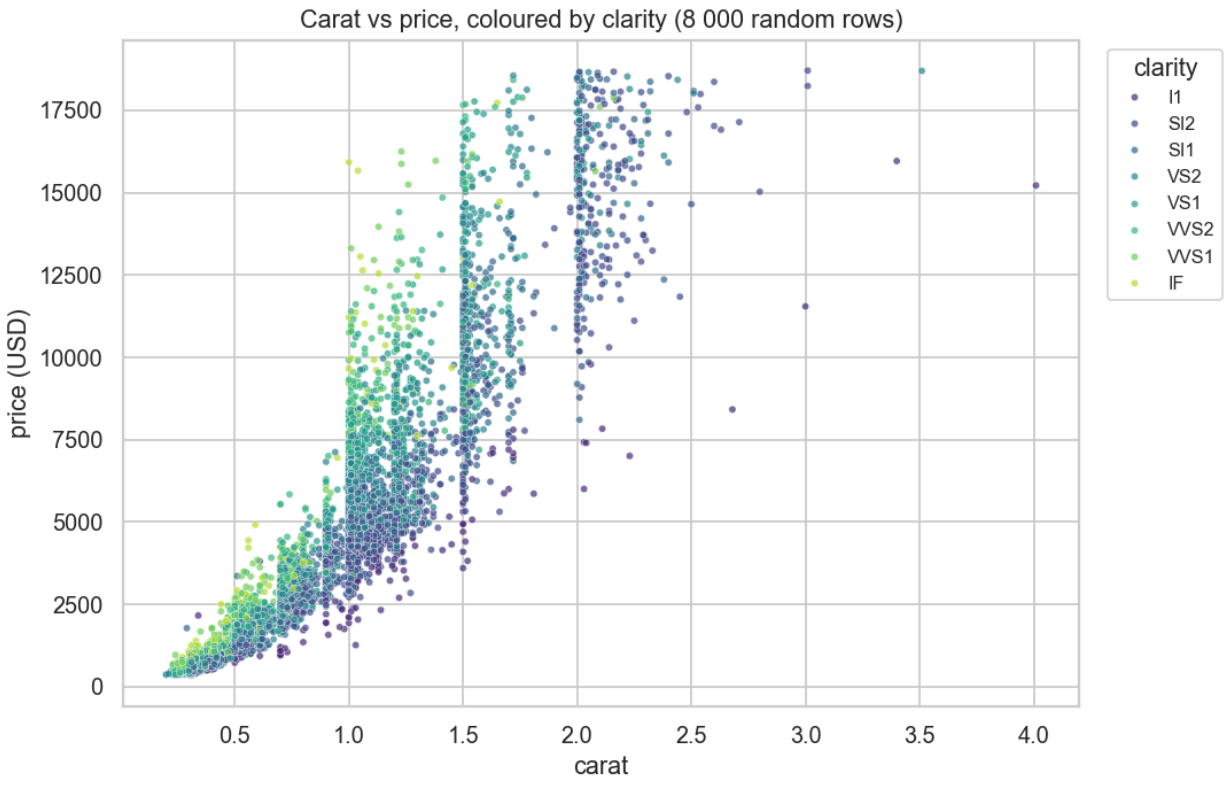


Figure 2: Sampled carat-price relationship colored by clarity. Higher carat generally increases price, but dispersion grows for larger stones, motivating nonlinear models and robust error analysis.

3 Related Work

The diamonds data set is documented in the ggplot2 reference as a 53,940-row data frame for prices and attributes of round-cut diamonds [1]. It is frequently used for exploratory regression because price depends on both continuous measurements and ordered quality grades.
Random Forests were introduced by Breiman as ensembles of randomized tree predictors whose aggregation improves generalization and provides variable-importance estimates [2]. The scikit-learn implementation describes RandomForestRegressor as an estimator that fits many decision-tree regressors on subsamples and averages their predictions to improve accuracy and control over-fitting [3]. Gradient tree boosting is represented here by XGBoost. Chen and Guestrin describe XGBoost as a scalable tree-boosting system designed for efficient sparse-aware learning and approximate tree construction [4]; the XGBoost Python API provides the scikit-learn-compatible XGBRegressor used in this project [5].
The neural-network comparison uses scikit-learn's MLPRegressor, which trains iteratively by updating parameters from loss gradients and can include regularization and early stopping [6]. The Adam optimizer used by the implementation is a first-order stochastic optimization method that estimates first and second moments of gradients [7]. The overall preprocessing and evaluation pipeline is based on scikit-learn estimators, ColumnTransformer, TransformedTargetRegressor, and cross-validation utilities [8, 9, 10].
Diamond-price-specific studies support the same modeling direction. A benchmark in Scientific Reports compared supervised learning algorithms for diamond price prediction and concluded that size variables, log transformations, and nonlinear models are central to this task. Under their specific evaluation protocol, the authors reported that a Random Forest regressor achieved the highest precision with an RMSE of 523.50 USD ($R^2 = 0.975$), outperforming their Multi-Layer Perceptron architecture, which yielded an RMSE of 563.74 USD ($R^2 = 0.980$). Notably, their implementation of XGBoost lagged significantly behind with an RMSE of 612.88 USD, leading them to select Random Forest as the superior model for this data family. While direct numeric comparison is limited by differences in the exact preprocessing and splitting steps, this baseline framework strongly justifies evaluating tree ensembles and neural regressors side by side.

4 Proposed Method

4.1 Preprocessing

The feature matrix uses six numeric predictors: carat, depth, table, x, y, and z, plus three ordered categorical predictors: cut, color, and clarity. Categorical grades are first encoded as ordinal integers following their natural domain order. The carat feature is transformed with $\log(1 + \text{carat})$, because carat is right-skewed and is the dominant size predictor. Numeric features are standardized for scale-sensitive models and passed through unchanged for tree-based models.
For the linear regression and MLP candidates, the target is transformed as

$$z = \log(p),$$

where p is price in USD. Predictions are mapped back to dollars with $\hat{p} = \exp(\hat{z})$. This log-target wrapper reduces the influence of extreme high-price rows while preserving final metrics on the original dollar scale. The wrapper follows the TransformedTargetRegressor pattern, where the regressor fits transformed targets and predictions are inverse-transformed for evaluation [9].

4.2 Model Families

Five candidate models are trained:

- **Mean baseline**: predicts the training-set mean price.
- **Linear regression**: scaled numeric features, ordinal categorical grades, and log-target training.
- **Random Forest**: 300 trees, maximum depth 15, minimum samples split 5, unscaled numeric features, ordinal categorical grades, and log-target training.
- **XGBoost**: 400 boosting rounds, learning rate 0.05, maximum depth 6, subsample 0.8. histogram tree method, ordinal categorical grades, and log-target training.

- **MLP regressor**: hidden layers $(64, 32)$, ReLU activation, Adam optimizer, L_2 regularization $\alpha = 10^{-2}$, batch size 256, early stopping with a 10% validation fraction, scaled numeric features, and one-hot categorical features.

The MLP uses one-hot categorical preprocessing rather than ordinal pass-through. This avoids implying equal numeric spacing between grade levels for a dense neural network, while the tree models retain ordinal encodings because tree splits can partition ordered codes naturally.

4.3 Evaluation Protocol

The cleaned data are split once into a training split and a held-out test split. Model selection uses only the training split. Five-fold shuffled K-fold cross-validation with random state 42 estimates RMSE, MAE, R^2, and MAPE. The primary selection metric is mean CV RMSE in dollars. After the best model family is selected, all candidate models are fitted on the full training split and evaluated once on the held-out split. This preserves the test set as a final estimate rather than a tuning signal.

For observations $i = 1, \ldots, n$, with true prices y_i and predictions $\hat{y}_i$, the main metrics are:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|.$$

R^2 measures explained variance, and MAPE reports relative error. RMSE is used for selection because large pricing errors on expensive diamonds are operationally important; MAE is reported for direct dollar interpretability.

5 Results and Discussion

5.1 Cross-Validated Results

Table 1 reports five-fold CV results on the training split. The mean baseline performs poorly, with RMSE 3971.57 USD and negative R^2, which confirms that the predictors carry substantial price information. The log-target linear model reduces RMSE to 1467.40 USD but remains far behind nonlinear methods. XGBoost obtains the lowest CV RMSE, 524.50 USD, followed by Random Forest at 548.49 USD and MLP at 576.9 USD.

Table 1: Five-fold cross-validation on the training split. RMSE and MAE are in USD. Lower RMSE, MAE, and MAPE are better; higher R^2 is better.

Model	RMSE mean ± std	MAE mean ± std	R^2 mean ± std	MAPE mean ± std
Mean baseline	3971.57 ± 42.24	3019.90 ± 16.51	-0.0002 ± 0.0001	1.8750 ± 0.0331
Linear regression	1467.40 ± 153.50	559.49 ± 12.35	0.8618 ± 0.0292	0.1272 ± 0.0007
Random Forest	548.49 ± 9.61	271.76 ± 3.66	0.9809 ± 0.0006	0.0651 ± 0.0007
XGBoost	**524.50 ± 8.22**	**261.78 ± 2.85**	**0.9826 ± 0.0004**	**0.0623 ± 0.0005**
MLP regressor	576.90 ± 37.83	289.08 ± 6.52	0.9788 ± 0.0032	0.0728 ± 0.0021

The ranking is stable across dollar and relative metrics: XGBoost is first, Random Forest second, MLP third, linear regression fourth, and the mean baseline last. The MLP result is important because it closes the final model set: a modest neural network improves dramatically over the linear baseline but does not surpass tree ensembles on this tabular data. This is consistent with the practical pattern that boosted trees often dominate medium-size structured data without extensive feature engineering.

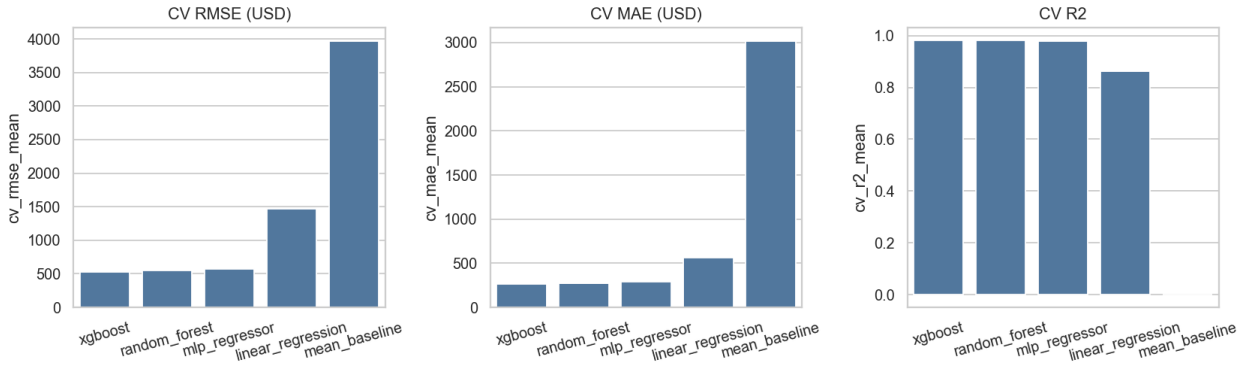


Figure 3: Cross-validated metric comparison. Nonlinear tree ensembles dominate the linear and mean baselines, while the MLP remains competitive but not best.

5.2 Held-Out Test Results

Table 2 reports the single held-out evaluation. XGBoost remains best, with RMSE 534.80 USD, MAE 269.49 USD, $R^2 = 0.9825$, and MAPE 0.0628. Random Forest is close, with RMSE 543.06 USD and $R^2 = 0.9819$. MLP achieves RMSE 564.25 USD and $R^2 = 0.9805$, confirming that it generalizes well but does not improve the final selection.

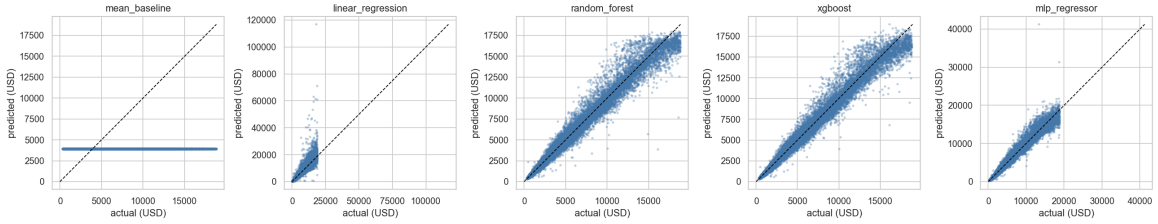
Table 2: Held-out test metrics. RMSE and MAE are in USD.

Model	RMSE	MAE	R^2	MAPE
Mean baseline	4042.03	3057.65	-0.0001	1.8770
Linear regression	1420.77	567.83	0.8764	0.1268
Random Forest	543.06	271.14	0.9819	0.0639
XGBoost	**534.80**	**269.49**	**0.9825**	**0.0628**
MLP regressor	564.25	291.74	0.9805	0.0737

The final test scores closely match CV estimates: XGBoost CV RMSE is 524.50 USD and held-out RMSE is 534.80 USD. This small gap suggests that the cross-validation selection did not overfit heavily to the training folds. The Random Forest and MLP also preserve their CV ordering on the held-out split.

5.3 Residual and Prediction Diagnostics

Figure 4 compares cross-validated predictions and residuals. The mean baseline collapses to a horizontal prediction band and is not a useful pricing model. Linear regression captures the main price trend but shows larger dispersion and systematic errors, especially at higher prices. The three nonlinear candidates produce points close to the identity line over most of the price range.



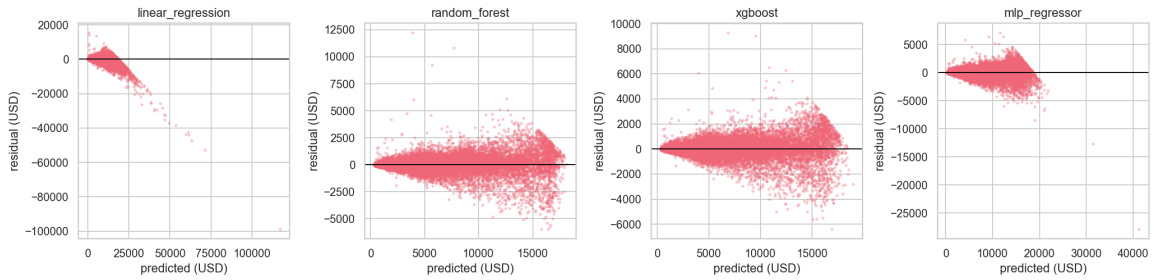


Figure 4: Cross-validated prediction and residual diagnostics. Nonlinear models track the identity line more closely than the linear and mean baselines, and residual spread remains largest for expensive stones.

Residual spread increases with predicted price, which is expected for a monetary target with higher absolute variance among larger diamonds. The log-target transformation reduces this effect but does not remove it completely. Remaining errors likely come from unobserved market factors, certification differences, temporal price variation, brand and seller effects, and cut details not captured by the nine predictors.

5.4 Learning Curves and Feature Importance

The learning and loss curves in Fig. 5 explain model behavior. The generated linear-regression learning curve showed similar train and validation RMSE at the largest training size, indicating high bias rather than overfitting. Adding more training data does not close the gap to the tree ensembles, which points to limited model capacity rather than insufficient sample size. This matches the EDA: the price relationship is nonlinear and depends on interactions between size, proportions, and quality grades. While initial gains in the Random Forest begin to flatten after 200 trees, marginal improvements continue until 300 estimators, which was selected as the optimal configuration by the cross-validation search. XGBoost's training curve converges smoothly on log-price RMSE, with validation error flattening after the main boosting phase. The MLP training loss declines while early-stopping validation R^2 stabilizes, indicating that the neural baseline learned a useful nonlinear representation but remained slightly less accurate than the tree ensembles.

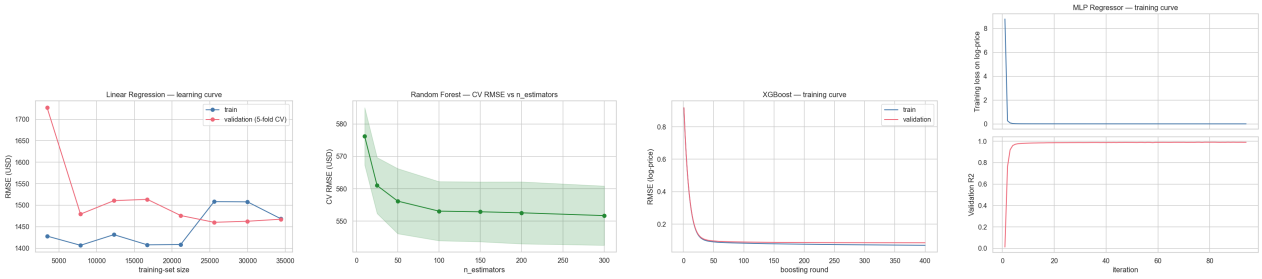


Figure 5: Representative learning and training curves. Linear regression remains bias-limited, Random Forest saturates as trees are added, XGBoost converges smoothly, and the MLP learns under early stopping but does not overtake the boosted trees.

Feature importances in Fig. 6 emphasize the same structure as the EDA: size dominates price prediction. Random Forest assigns the largest importance to y, followed by carat, x, clarity, and color. XGBoost also emphasizes y, carat, z, and x, with smaller but nonzero contributions from clarity and color. The high importance of one dimension should not be interpreted as a causal claim that width alone determines price; it reflects collinearity among carat and the physical dimensions. The tree models can use one correlated size proxy heavily while still benefiting from the broader feature set.

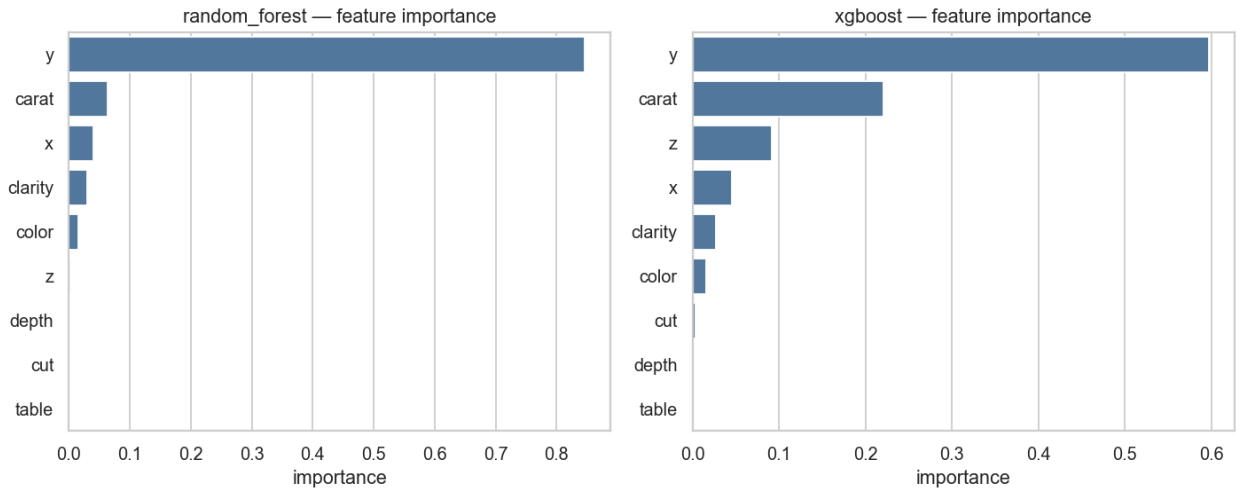


Figure 6: Random Forest and XGBoost feature importance. Size-related variables dominate, with clarity and color contributing additional quality information.

6 Reproducibility and Code Quality

The repository is organized into reusable source modules, scripts, reports, models, tests, and data. The core package contains data validation and splitting, preprocessing, model factories, and metric helpers. The scripts regenerate training tables, plots, diagnostics, loss curves, and saved-model inference checks. Tests cover data loading, cleaning, split behavior, preprocessing shapes, model fit/predict smoke checks, and positive finite predictions. The complete experiment can be run from the repository root by setting PYTHONPATH=src and executing the EDA, training, loss-curve, diagnostics, final-evaluation, and prediction scripts in sequence. The unit test suite uses Python unittest discovery under the tests directory. The final regenerated run completed 12 tests successfully. The saved model checkpoint and JSON metadata record the selected model, selection metric, CV scores, and held-out metrics. Several implementation choices reduce leakage risk. Preprocessing is kept inside model pipelines, so scaling and encoding are fit within each training fold. The log-target transform is wrapped around the estimator and inverted only for prediction. The held-out split is not used for model selection. The final metrics table is produced only by the explicit final-evaluation command.

7 Conclusion

The completed model comparison selects XGBoost as the final diamond price predictor. It achieves the best five-fold CV RMSE and the best held-out test performance, with final RMSE **534.80** USD, MAE **269.49** USD, $R^2 = \mathbf{0.9825}$, and MAPE **0.0628**. Random Forest is a close second, while the MLP baseline is clearly useful but not superior to the tree ensembles. Linear regression remains valuable as an interpretable sanity check, but it cannot capture enough nonlinear structure for best accuracy. The main limitation is that the data contain only static physical and quality variables. Real market prices also depend on certification, retailer, time, location, shape variations, and supply-demand dynamics. The feature-importance analysis is also affected by collinearity among carat and dimensions. Future work should evaluate calibrated prediction intervals, monotonic constraints for domain consistency, richer hyperparameter optimization, and external validation on a temporally or commercially distinct diamond-pricing data set.

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